

INSPECT-SR Guidance

INveStigating ProblEmatic Clinical Trials in Systematic Reviews

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Version 1.1.2

Table of contents

Preface	2
How to use this document	2
Domain 1: Inspecting post-publication notices	3
Domain 1: Inspecting post-publication notices	4
Check 1.1: Retraction status	5
Rationale	5
How to perform the check	5
Interpretation	5
Check 1.2: Expressions of concern	6
Rationale	6
How to perform the check	6
Interpretation	6
Domain 2: Inspecting conduct, governance and transparency	7
Domain 2: Inspecting conduct, governance and transparency	8
Check 2.1: Prospective registration	9
Rationale	9
How to perform the check	9
Appendices	10
Changelog	11
Version 1.1.2	11
Version 1.1.1	11

Preface

This guidance supports the use of INSPECT-SR, a tool for identifying problematic randomised controlled trials in systematic reviews of healthcare interventions. Each check is presented in a consistent structure so that reviewers can apply them independently and reproducibly.

i Scope of this guidance

INSPECT-SR is intended for use on randomised controlled trials that have been included, or are being considered for inclusion, in a systematic review.

How to use this document

Each check is described under four headings: rationale, how to perform the check, how to interpret the result, and worked examples. Checks are grouped into domains.

Table 1: Overview of the four INSPECT-SR domains

Domain	Focus	Checks
1	Post-publication notices	3
2	Conduct, governance and transparency	5
3	Text and figures	2
4	Results in the study	11

See Table 1 for the domain structure.

Domain 1: Inspecting post-publication notices

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Check 1.1: Retraction status

Rationale

Retracted trials should not contribute to the evidence base. Retraction notices are inconsistently indexed, so a positive check requires searching more than one source.

How to perform the check

1. Search the Retraction Watch Database for the trial.
2. Check the journal's own website for a retraction or expression of concern.
3. Check PubMed for a linked comment or correction.

Indexing lag

Retraction notices can take months to propagate to bibliographic databases. A negative result is not conclusive.

Interpretation

If a retraction is found, exclude the trial and record the reason.

```
library(dplyr)
trials |>
  filter(!is.na(retraction_date)) |>
  select(trial_id, journal, retraction_date)
```

Check 1.2: Expressions of concern

Rationale

An expression of concern signals unresolved doubt about a trial's integrity.

Expressions of concern are issued when an editor has well-founded concerns but the investigation is incomplete.

How to perform the check

Search the same sources as Check 1.1, recording the date and the stated reason.

Interpretation

Treat as a signal for further scrutiny rather than automatic exclusion.

Domain 2: Inspecting conduct, governance and transparency

Domain 2: Inspecting conduct, governance and transparency

Check 2.1: Prospective registration

Rationale

Absence of prospective registration is associated with selective reporting.

Practical tip

Compare the registration date with the stated first-enrolment date, not the publication date.

How to perform the check

Locate the registration record and compare dates.

Important

Discrepancies of more than one month warrant a query to the authors.

Appendices

Changelog

Version 1.1.2

- Clarified the wording of Check 2.1.
- Added worked examples to Domain 4.

Version 1.1.1

- Initial public release.